Viacheslav Kruglov

Computational Physicist · Scientific Software Developer · Numerical Algorithms
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PROFILE

C++ and Python developer with a PhD in physics and mathematics and about ten years in numerical and simulation software, including processing of large datasets. Develop and maintain industrial C++ control software; built a parallel particle transport framework under memory constraints.

SKILLS

Core: C++20, C++17, Python, Linux, Bash; MATLAB (university teaching)
Parallel computing: Intel TBB, OpenMP, SLURM, shared memory parallelism, processing with bounded memory
Numerics: ODE/PDE models, integration and interpolation, subspace analysis with SVD, eigenvalue problems, LAPACK, FFTW
Data processing: NumPy, pandas, NetCDF, zlib
Build and testing: CMake, Git, Boost.Test, CTest, GitLab CI, GitHub Actions
Development: Library and interface design, object-oriented C++, Visual Studio, GDB, Valgrind, Windows; Scrum, Jira, Confluence

PROFESSIONAL EXPERIENCE

Software Developer 05/2026-present
Big Dutchman Service GmbH, Vechta

- Maintain and extend C++ control software, including stir fan control in the climate control system; add Boost.Test unit tests for changes.
- Extend C++ components based on abstract classes while preserving existing interfaces.
- Developed test tooling with a scanner simulator for a livestock monitoring system, so automated tests could run without scanner hardware.
- Documented subsystem behaviour and known issues in Confluence to support troubleshooting and maintenance.

Postdoctoral Researcher / Research Software Developer 01/2023-02/2026
Carl von Ossietzky University of Oldenburg, ICBM, Oldenburg

- Developed a C++20/CMake framework for particle transport in unsteady ocean currents under Linux, combining spherical integration, kd-tree queries and spline interpolation.
- Parallelised particle propagation with Intel TBB on the university HPC cluster (SLURM); production runs were configured for 2.8 million particles per model day.
- Kept memory use bounded during runs lasting several days by loading rolling time windows of NetCDF data.
- Tested a prototype against an analytical flow field and checked time step sensitivity before analysing transport statistics in realistic fields.

Research Scientist, Laboratory of Theoretical Nonlinear Dynamics 10/2017-10/2022
Kotelnikov Institute of Radioengineering and Electronics (RAS), Saratov Branch

- Developed C++ and Python models of oscillator, chemical and mechanical systems described by ODEs and PDEs.
- Computed Lyapunov exponents and tested hyperbolicity using SVD, eigenvalue analysis and tangent space methods with LAPACK; analysed spectra with FFTW.
- Ran studies of roughly one million parameter combinations with OpenMP and investigated regimes requiring long integrations.

Software Engineer, Industrial Simulation 02/2017-02/2018
INACOS (INIUS JSC), Saratov

- Implemented models of thermodynamic properties and integrated a vacuum distillation column model into an industrial simulation platform.
- Compared calculated process quantities with documented reference values and corrected discrepancies in the model and implementation.

PUBLIC REPOSITORY

particle-transport-framework
github.com/chaotic-coconut/particle-transport-framework

- Published the framework as a header-only C++20/CMake library that builds with both the old and the new Boost spline API and with different netCDF installations.
- Maintain CTest checks for coordinate transforms, great circle distance derivatives, date calculations and compressed record round trips, with AddressSanitizer and UBSan builds in GitHub Actions.

RESEARCH AND PUBLICATIONS

Researched nonlinear dynamics and chaotic systems, including numerical verification of hyperbolicity.
Published 20 journal articles, four in *Chaos*, and two conference papers.

TEACHING AND TALKS

- Gave lectures and seminar classes in Synchronisation and Statistical Radiophysics, and physics classes, at Saratov State Technical University (part time, 2018-2019 and 2020-2022).
- Designed and taught beginner Python courses and modelling practicals at the University of Oldenburg, including assessment.
- Presented a talk at Dynamics Days Europe 2025 and posters at Dynamics Days Europe 2024 and the EGU General Assembly 2025 in Vienna.

EDUCATION

PhD in Physics and Mathematics 2012-2016
Saratov State University, Saratov
Thesis: *Finite-dimensional and distributed ring-structured systems generating robust chaos.*
Assessed by the German ZAB as equivalent to a German doctorate.

Diploma in Physics, with honours 2007-2012
Saratov State University, Saratov
Five years of study. Specialisation: Physics of Open Nonlinear Systems within Computer Methods of Physics.

ADDITIONAL INFORMATION

Languages: English (fluent, working language), German (Goethe-Zertifikat B1), Russian (native)
Work authorisation: Permanent residence permit (Niederlassungserlaubnis); no sponsorship required